Introduction to Business Intelligence & Power BI

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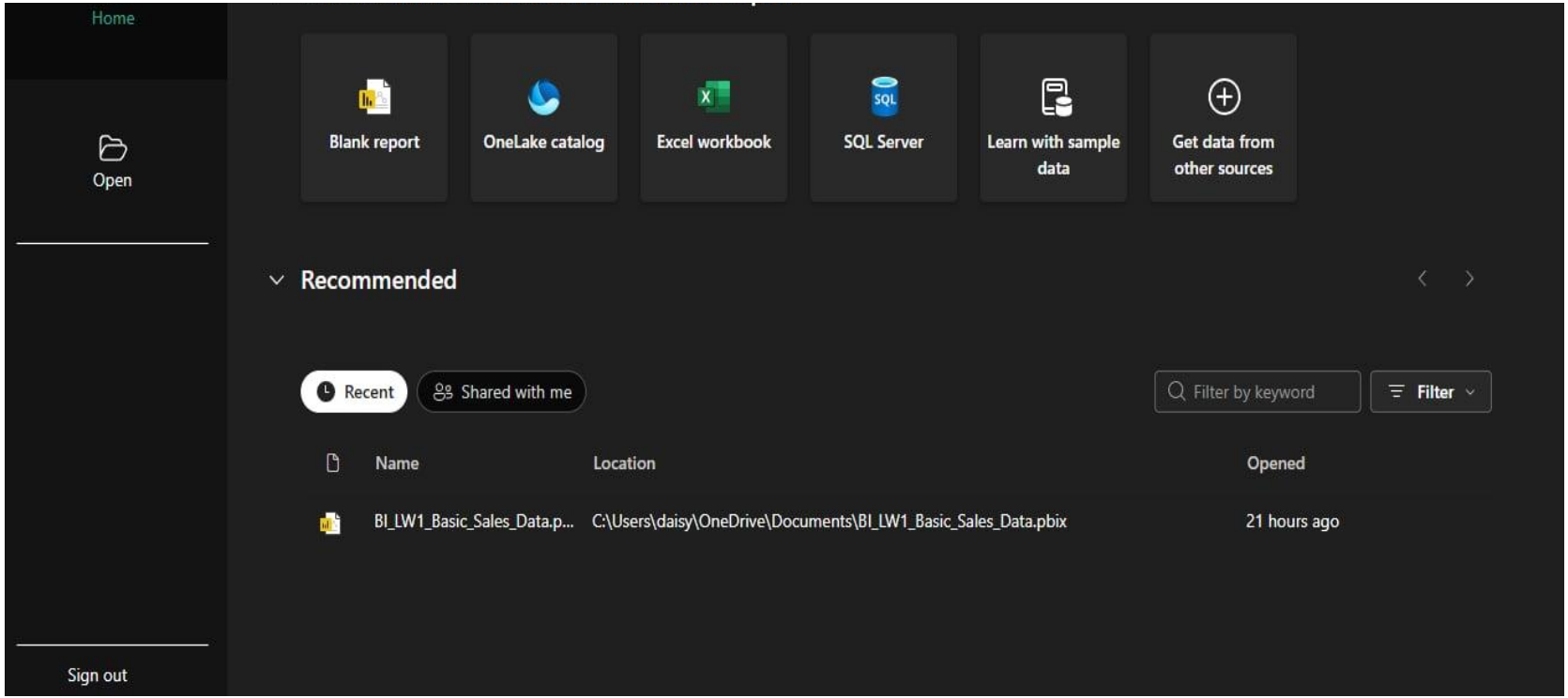
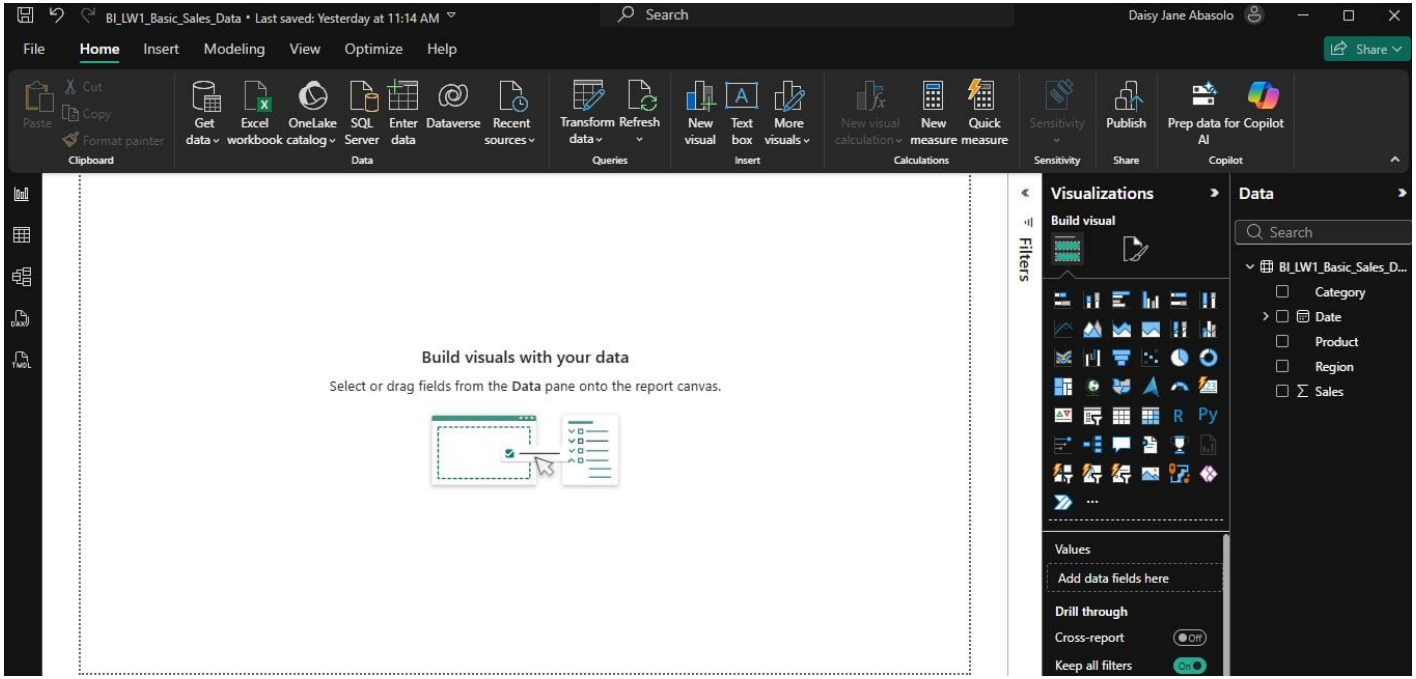
IS 107 - Business Intelligence

Step 1: Open Power BI Desktop



Step 2: Load the Dataset



Step 3: Verify Data in Data View

1. Click the Data View icon (table icon on the left)
2. Check:

○ Are all columns visible?

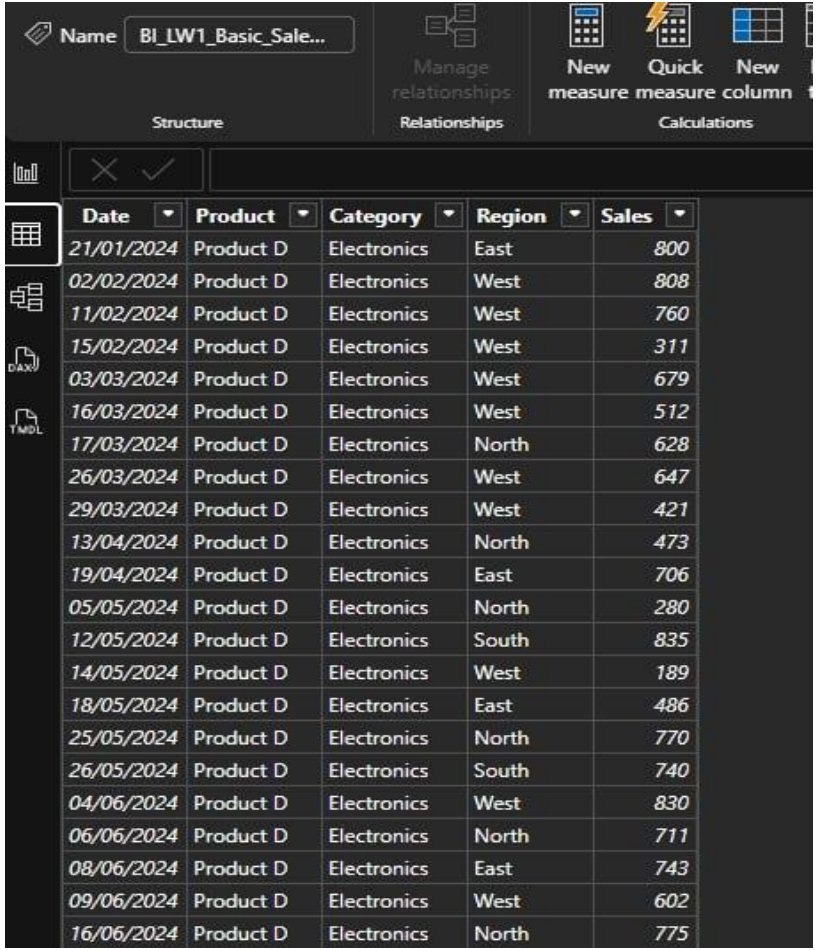
Yes, all columns are visible in Power BI's Data View.

○ Is "Date" formatted as Date?

Yes. The "Date" column is already formatted as Date in my Power BI dataset

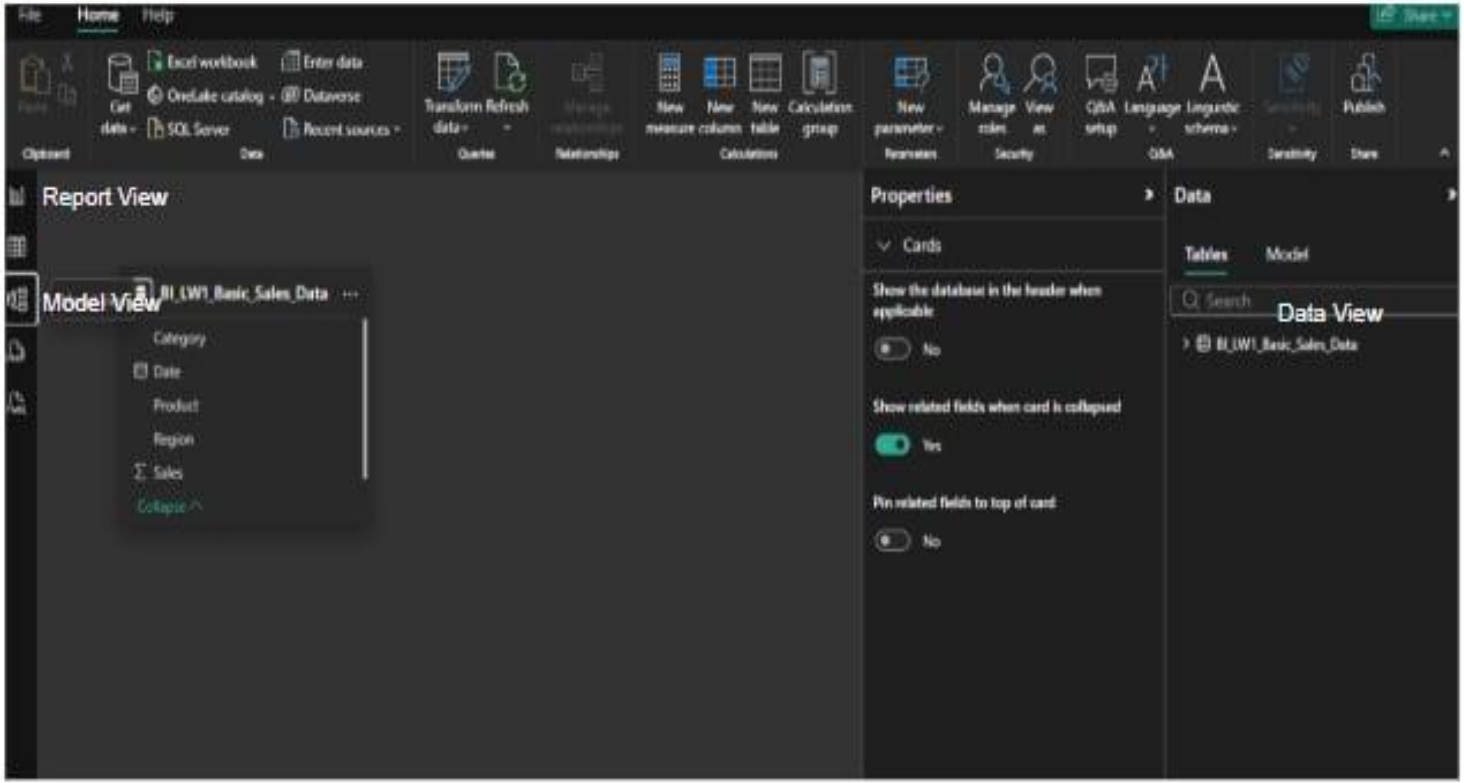
○ Is "Sales" formatted as Decimal Number?

Yes. The "Sales" column is formatted as a Decimal Number in Power BI,which allows the visuals to correctly calculate totals, sums, and other aggregations, like showing $220.23K in the charts.

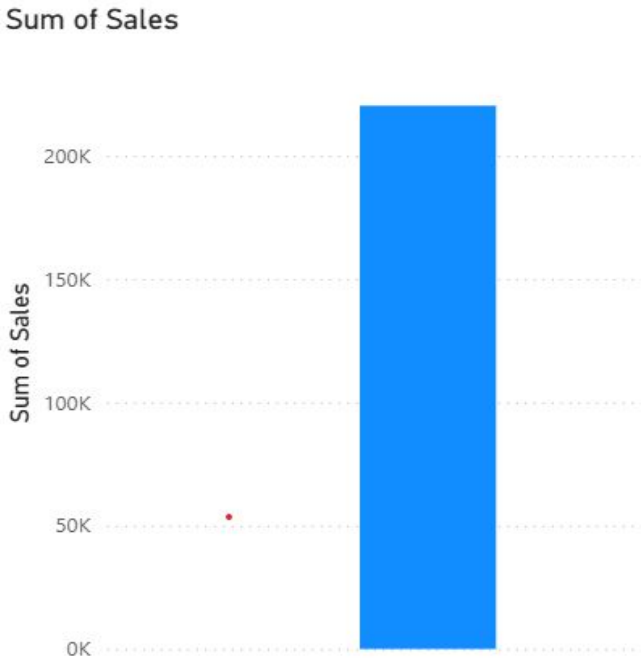


Date	Product	Category	Region	Sales
21/01/2024	Product D	Electronics	East	800
02/02/2024	Product D	Electronics	West	808
11/02/2024	Product D	Electronics	West	760
15/02/2024	Product D	Electronics	West	311
03/03/2024	Product D	Electronics	West	679
16/03/2024	Product D	Electronics	West	512
17/03/2024	Product D	Electronics	North	628
26/03/2024	Product D	Electronics	West	647
29/03/2024	Product D	Electronics	West	421
13/04/2024	Product D	Electronics	North	473
19/04/2024	Product D	Electronics	East	706
05/05/2024	Product D	Electronics	North	280
12/05/2024	Product D	Electronics	South	835
14/05/2024	Product D	Electronics	West	189
18/05/2024	Product D	Electronics	East	486
25/05/2024	Product D	Electronics	North	770
26/05/2024	Product D	Electronics	South	740
04/06/2024	Product D	Electronics	West	830
06/06/2024	Product D	Electronics	North	711
08/06/2024	Product D	Electronics	East	743
09/06/2024	Product D	Electronics	West	602
16/06/2024	Product D	Electronics	North	775

PART 2: Exploring the Interface



PART 3: Creating Auto-Generated Visuals
Step 1: Quick Visualization



Question:

- What type of chart was created?

This is a Bar Chart. .

- What does it show?

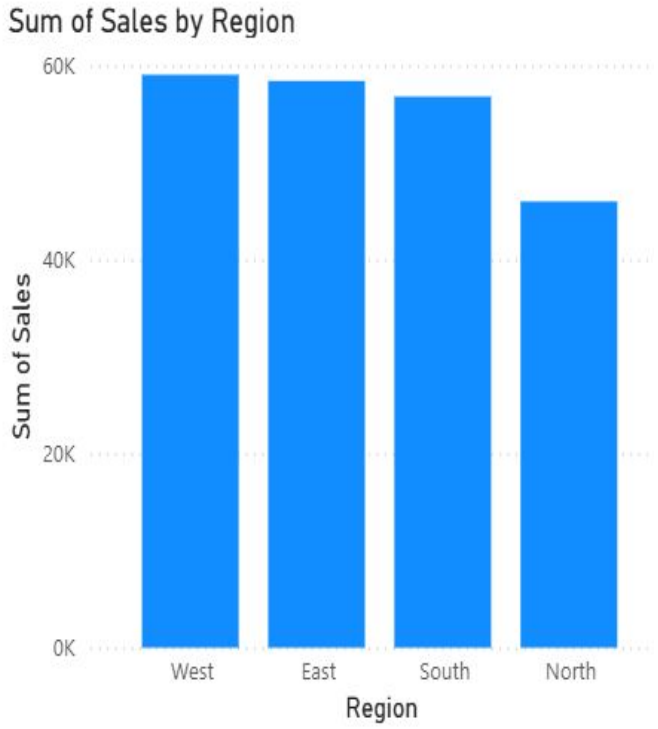
It shows the total sales for each category or region. Specifically,
it represents the “Sum of Sales”, meaning the total amount of money generated,
allowing you to easily compare which categories or regions perform better.

Step 2: Create a Sales by Region Chart

Question:

- Which region has highest sales?

The West region has the highest sales



Step 3: Sales by Category

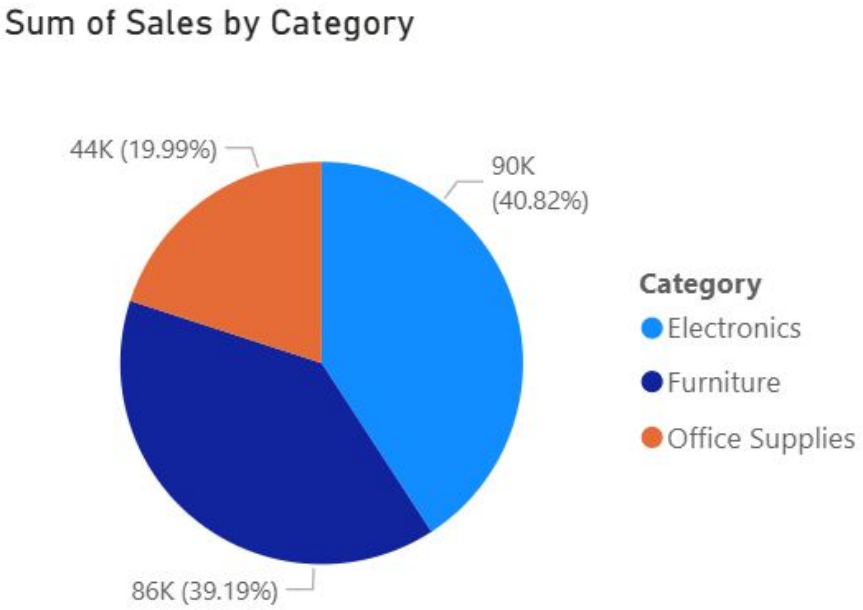
Question:

- Which category dominates?

The Electronics category dominates, with the highest sales of 90K, making up 40.82% of the total sales.

- Is the distribution balanced?

No, the distribution is not balanced. While Electronics (40.82%) and Furniture (39.19%) are almost equal, Office Supplies only makes up 19.99% of sales, which is about half of the other two categories, making the overall distribution uneven.



Step 4: Sales Over Time

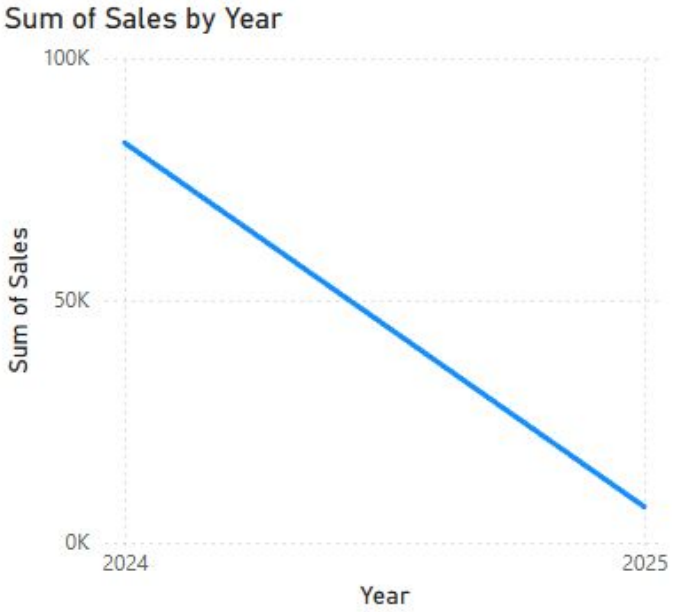
Question:

- Is there growth?

There is no growth. Sales went down a lot between the two years.

- Any noticeable trend?

Yes, there is a noticeable trend sales are decreasing sharply over time. In the Sales Over Time chart, sales started high in 2024 at around 80K but dropped quickly in 2025 to just above 10K, showing a significant decline year over year.



PART 4: Basic Data Insight Interpretation

Question:

● Which region contributes most revenue?

The West region contributes the most revenue, totaling approximately $59,000.

● Which product category performs best?

The Electronics category performs the best, making $90,000 in sales, which is 40.82% of the total.

● Are sales consistent across dates?

No, sales are very inconsistent.

● What business recommendation can you suggest?

Investigate why sales dropped almost 87% in 2025. Focus marketing on the West and East regions, and on Electronics and Furniture, since these bring in the most sales.

LABORATORY QUESTIONS

Part A – Technical Questions

1. What are the five columns in the dataset?

Region, Category, Year, Sales, and usually a fifth column like Date or Product Name.

2. What data type is assigned to the “Sales” column?

Decimal Number.

3. Which Power BI view allows you to see raw data?

Data View.

4. What chart type is best for showing trends over time?

Line chart.

5. What aggregation is automatically applied to Sales?

Sum aggregation.

Part B – Analytical Questions

6. Which region has the highest total sales?
West region, nearly $60,000.

7. Which category has the lowest performance?
Office Supplies, 19.99% ($44K) of total sales.

8. Are sales increasing, decreasing, or stable?

Decreasing from about $80,000 in 2024 to around $10,000 in 2025.

9. If you were a manager, which region would you prioritize?
North region for growth ($46K), representing the largest opportunity.

10. Provide one actionable recommendation based on the data

Investigate the 2025 sales drop of ~87% to prevent potential business issues.

Task 1: Add a Card Visualization

Question:

- What is the total sales amount?

  220.23K

Task 2: Add Slicer

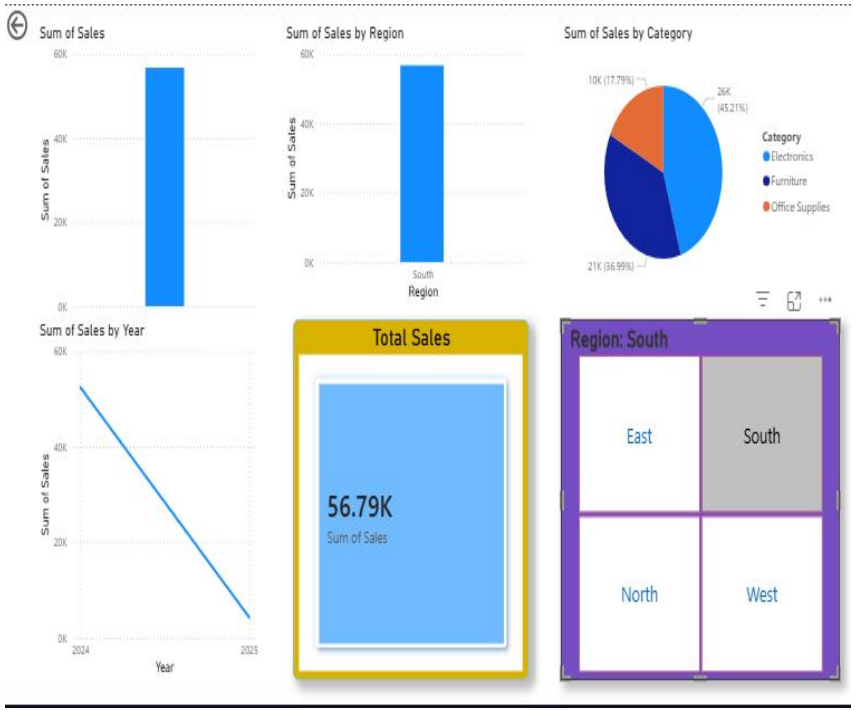
Question:

- What happens to other visuals when you click a region?

All other charts update to show only the selected region's data.

- Why is filtering important in BI?

Filtering is crucial in Business Intelligence (BI)
because it helps you focus on the most relevant data
and extract meaningful insights without being overwhelmed
by the full dataset.



Task 3: Sort Sales

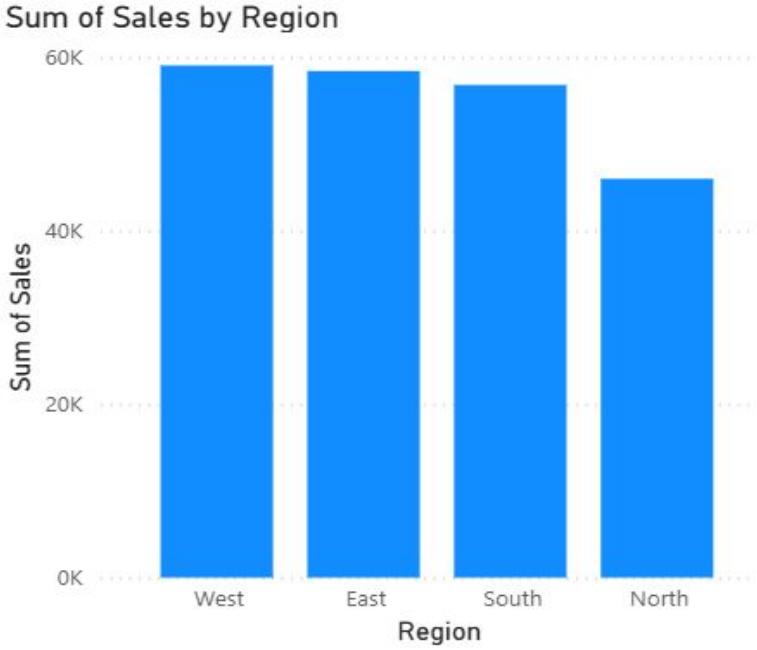
Question:

- Does sorting improve readability?

Yes, sorting improves readability because it makes trends obvious, allows easier comparisons, reduces cognitive load, highlights outliers, and helps communicate insights clearly.

- Why?

Sorting improves readability because it organizes data in a logical order, making it easier for the viewer to understand patterns, compare values, and quickly identify important information without having to search through random or unsorted data.



Task 4: Identify Outliers

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- Which region is significantly higher or lower?

West ($59K) and East ($58K) are higher; North ($46K) is lower.

- What might explain that difference?

West and East may have larger markets, stronger demand, or better supply chains; North may have fewer stores, smaller teams, or supply issues.